

Aurua

SUBMISSION PACKET — PHASE 3

PROJECT	Aurua — agent-driven explainer video generation, grounded in source material
TRACK	A — Technical Build
AUTHOR	Jorge Urias (individual project)
COURSE	94815 Agentic Technologies · CMU Heinz College · Prof. Anand S. Rao
PHASE	3 of 3
SUBMITTED	April 29, 2026
REPOSITORY	https://github.com/Misterurias/Aurua-Video-Generation
5-MINUTE VIDEO	See <code>media/demo_video_link.txt</code> in the repository

Project Summary

Aurua takes a student's question about a confusing topic in 3Blue1Brown's neural-network video series and produces a short animated explainer video, generated end-to-end from a verified scene plan to rendered Manim output. The system uses four LLM agents (Intent & Grounding, Planner, Verifier, Animation Coder) and a deterministic Renderer tool, with a verifier-driven retry loop and a parallel branch for animation code generation. Across 12 evaluation runs spanning two architecture versions, the canonical Phase 3 demo run (`d79b8b7209a2`) produces a verified, grounded, ~80-second silent rendered video for a custom student question — first-try verifier approval, all four scenes grounded with confidences 0.82–0.88, total cost \$0.16, total wall-clock 94 seconds.

Headline result. End-to-end pipeline working: question → grounded scene plan → generated Manim code → rendered video. The verifier loop catches real planner overreach (documented across runs `170daccbae47` and `97fb7e6b6768`); the system correctly exhausts on questions the source can't ground rather than ship hallucinated content. Five distinct failure modes (F1–F5) are documented honestly with specific paths forward.

Submission Package Contents

The repository is organized so a reviewer can find any artifact within five minutes.

Top-level documents

DOCUMENT	PATH	PURPOSE
README	<code>README.md</code>	Project overview, setup, run instructions, evaluation summary
Final report (this is the polished read)	<code>docs/final_report.pdf</code>	14-page report with executive summary, architecture, evaluation, failure analysis, governance, lessons learned
Phase 3 document (comprehensive technical record)	<code>docs/PHASE_3.md</code>	Full technical record mirroring the Phase 2 document's structure, with grader-feedback response and individual contribution reflection
Architecture diagram	<code>docs/architecture_diagram.pdf</code>	Single-page diagram showing all 4 agents + Renderer + verifier loop + Phase 3 rendering branch
AI usage disclosure	<code>AI_USAGE.md</code>	Eight session entries covering Phase 1 → Phase 3, with verification notes
Phase 2 document (for context)	<code>docs/PHASE_2.md</code>	Previous phase document

Evaluation package

ARTIFACT	PATH	DESCRIPTION
Test cases	<code>eval/test_cases.csv</code>	5 scenarios + 1 ablation, defined in Phase 2
Evaluation results	<code>eval/evaluation_results.csv</code>	All 12 runs across v0.2 and v0.3
Failure log	<code>eval/FAILURE_LOG.md</code>	5 documented failures (F1-F5) with verifier quotes and trace pointers
Version notes	<code>eval/VERSION_NOTES.md</code>	v0.2 baseline, v0.3 changes, v0.4 plan
Phase 2 archived runs	<code>eval/runs_saved/phase2_evaluation/</code>	6 archived run directories with traces and state.json
Phase 3 archived runs	<code>eval/runs_saved/phase3_evaluation/</code>	6 archived run directories including the canonical full-pipeline demo
Live trace files	<code>eval/traces/</code>	Per-call prompt/response logs for every run

Final artifact

ARTIFACT	PATH
Canonical Phase 3 demo (rendered video)	outputs/runs/d79b8b7209a2/final_silent.mp4
Per-scene MP4s	outputs/runs/d79b8b7209a2/videos/scene_01.mp4 through scene_04.mp4
Generated Manim code per scene	outputs/runs/d79b8b7209a2/scenes/scene_01.py through scene_04.py
Full state	outputs/runs/d79b8b7209a2/state.json
Full traces	eval/traces/d79b8b7209a2/

Visual evidence

ARTIFACT	PATH
Screenshots (8 captioned images)	docs/screenshots/01_*.png ... 08_*.gif
Screenshot index	docs/screenshots/screenshot_index.md

Source code

COMPONENT	PATH
Orchestrator	src/run.py
Shared state (Pydantic models)	src/state.py
Anthropic SDK wrapper	src/claude_client.py
Configuration	src/config.py
Agent 1 — Intent & Grounding	src/agents/intent_grounding.py
Agent 2 — Planner	src/agents/planner.py
Agent 3 — Verifier	src/agents/verifier.py
Agent 4a — Animation Coder	src/agents/animation_coder.py
Agent 4b — Narration Writer (scaffolded, v0.4)	src/agents/narration_writer.py
FAISS retrieval tool	src/tools/retrieval.py
Renderer tool	src/tools/renderer.py
System prompts	src/prompts/
TC6 single-prompt baseline	scripts/run_tc6_baseline.py

Evaluation Summary

12 evaluation runs total: 6 from Phase 2 (v0.2) and 6 from Phase 3 (v0.3).

TEST	TYPE	V0.2 RESULT	V0.3 RESULT	OUTCOME
TC1	Happy-path conceptual	verified (1 retry)	verified first try + verified after 1 retry on a separate run + canonical demo with full render	Improved
TC2	Primary mechanical anchor	exhausted (F1)	exhausted (F4)	Different failure mode
TC3	Out-of-scope	clarification_required ✓	(corpus-independent)	Correct refusal
TC4	Vague input	clarification_required ✓	(corpus-independent)	Correct refusal
TC5	Multi-path mechanical	exhausted (F2)	exhausted (F5)	Different failure mode
TC6	Single-prompt ablation	grounded 4/4	identical 4/4 (reproducibility check)	Stable baseline

Canonical Phase 3 demo. Run d79b8b7209a2 , executed 2026-04-29 against a custom student question on data/transcripts/3b1b_ch3.txt . Verified on first attempt (0 retries), 4 scenes rendered, ffmpeg-concatenated to final_silent.mp4 . **8 API calls, \$0.16, 94 seconds wall-clock total.** Archived at outputs/runs/d79b8b7209a2/ .

Documented failures (F1-F5)

5 failure cases — more than the rubric’s ≥ 2 requirement. Together they document a complete iteration cycle: identify failure → propose specific fix → run it → document what improved and what new issues surfaced → plan next iteration.

ID	VERSION	SEVERITY	STATUS	RESOLUTION
F1	v0.2	High	Partially closed in v0.3	Retrieval improved measurably; deeper finding (F4) emerged
F2	v0.2	Medium-high	Partially closed in v0.3	Retrieval improved; deeper finding (F5) emerged
F3	v0.2	n/a (architectural finding)	Documented; reproduced in v0.3	Architectural scope statement
F4	v0.3	Medium	Documented	Source informality limits formal grounding even with perfect retrieval; v0.4 source-augmentation planned
F5	v0.3	Medium	Documented	Smaller chunks introduced sentence-truncation artifacts; v0.4 sentence-boundary chunking planned

Full writeup with verifier quotes, run-state pointers, and trace references in `eval/FAILURE_LOG.md`.

Cost discipline

No run in the evaluation set exceeded 10 API calls or \$0.50, including full-rendering runs. The 2-retry verifier cap holds the cost ceiling regardless of input.

RUN TYPE	API CALLS	COST	WALL-CLOCK
Vague input early-stop (TC4)	1	\$0.013	3.1 s
Successful happy-path (TC1 v0.3 first try)	4	\$0.07	30.5 s
Successful with retry (TC1 v0.3 retry 1)	6	\$0.08	39.5 s
Canonical demo (TC1 v0.3 full pipeline, custom question)	8	\$0.16	94.0 s
Verifier exhaustion (TC2 v0.3)	8	\$0.10	50.7 s
Single-prompt baseline (TC6)	1	\$0.04	12-17 s

Reproducibility

The repository can be cloned and run end-to-end with three commands.

Setup

```
git clone https://github.com/Misterurias/Aurua-Video-Generation.git
cd Aurua-Video-Generation
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
brew install pkg-config cairo pango ffmpeg           # macOS Manim system deps
cp .env.example .env                                # add ANTHROPIC_API_KEY
```

Re-run the canonical demo

```
rm -rf data/index/                                     # ensure FAISS index rebuilds for Ch3
python -m src.run \
  --question "What does the gradient of the cost function tell us about which weights to change?" \
  --source data/transcripts/3b1b_ch3.txt
```

Expected: ~3-4 minutes wall-clock, ~\$0.15-0.50 cost (depending on whether the verifier needs to retry), final video at `outputs/runs/<run_id>/final_silent.mp4`.

Replay any archived run

Every archived run includes its full state and trace files. To inspect, e.g., the canonical demo's verifier output:

```
cat eval/traces/d79b8b7209a2/03_verifier_attempt_0.txt
```

Determinism: agents 1-3 use Sonnet at temperature 0.0 and produce reproducible output; the Animation Coder uses Opus 4.7 (which deprecated the temperature parameter mid-project) and is reproducible up to the model's defaults.

Pointers to Required Rubric Elements

For a grader walking the rubric category by category, here is where each piece of evidence lives.

RUBRIC CATEGORY	POINTS	WHERE TO LOOK
Final artifact quality and completeness	20	outputs/runs/d79b8b7209a2/final_silent.mp4 ; screenshots 02, 06, 07, 08
Evidence of agentic behavior or coordination	15	docs/architecture_diagram.pdf ; eval/traces/d79b8b7209a2/ ; screenshots 02, 03, 04, 05
Evaluation quality and results	15	eval/evaluation_results.csv (12 runs); docs/final_report.pdf § 5
Failure analysis and iteration	15	eval/FAILURE_LOG.md (F1-F5 with v0.2 → v0.3 fix-and-retest cycle); docs/final_report.pdf § 6
Governance, trust, and responsible behavior	10	docs/final_report.pdf § 7; verifier traces (screenshots 03, 04); state.json audit trail (screenshot 05)
Presentation (5-minute video)	10	media/demo_video_link.txt
Documentation, reproducibility, portfolio quality	15	README.md ; docs/PHASE_3.md ; AI_USAGE.md ; this packet

Honest Limitations and v0.4 Plan

Phase 3 stops short of three things the Phase 2 grader noted; each is documented honestly rather than papered over:

1. **Audio narration.** The Renderer produces silent video. Agent 4b (Narration Writer) is scaffolded but not wired; TTS and ffmpeg muxing are reserved for v0.4. The grader's stated minimum was "even if TTS and ffmpeg muxing are not complete" — this iteration meets that bar with a full silent video rather than just a single scene.
2. **Per-scene citation overlays in rendered video.** Designed in Phase 2 risk plan; not implemented. Reserved for v0.4. The audit trail still exists (state.json + trace files), just not as in-video text overlays.
3. **Per-agent typed views on** `RunState` . The Phase 2 grader noted no explicit access boundary. v0.3

documents the convention-enforced boundary (each agent's function signature shows what it reads); typed views are reserved for v0.4.

Full v0.4 priority list with effort estimates in `docs/PHASE_3.md` § 10 and `eval/VERSION_NOTES.md`.

Submission File Manifest

Files included in this submission, in addition to the repository link above:

1. `submission_packet.pdf` (this document) — primary Canvas submission
2. `docs/final_report.pdf` — polished final report
3. `docs/architecture_diagram.pdf` — Phase 3 architecture diagram
4. Repository link: <https://github.com/Misterurias/Aurua-Video-Generation>
5. 5-minute video link: in `media/demo_video_link.txt` within the repository

All other artifacts (source code, evaluation files, traces, screenshots, run states, generated Manim code, rendered videos) are accessible through the GitHub repository.